Chapter 5

VISUAL STIMULI IN COMMUNICATION

SPEECHREADING AND MANUAL COMMUNICATION

What It’s Like to Read Lips

https://www.youtube.com/watch?v=n1jLkYyODsc

Terminology

Lipreading: Using only the visual speech signal and related facial and body gestures to recognize speech.

Speechreading: Using the visual speech signal and the auditory signal and related facial and body gestures to recognize speech.

Some people use these terms interchangeably.

Speechreading for communication

People with perfectly normal hearing often rely on speechreading (i.e. watching a talker's face while also listening).

When someone attempts to recognize speech through visual signals, it activates the auditory cortex.

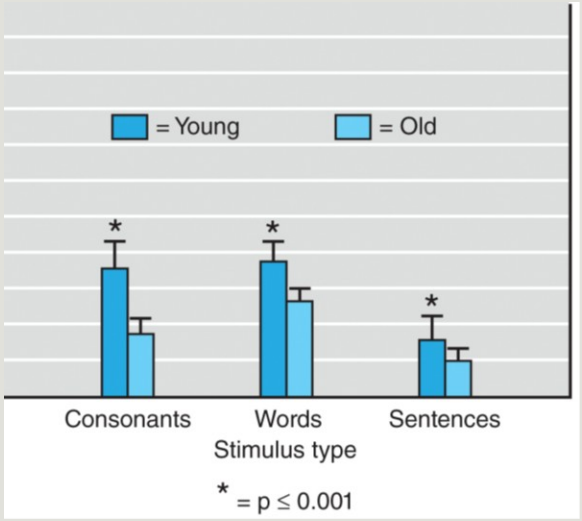
Sensitivity to the visual speech signal emerges at a very early age.



Predicting who will be a good lipreader

- **Variables that DO NOT have predictive power:**
  - Intelligence
  - Educational achievement
  - Presence of/duration of acquired hearing loss
  - Age at hearing loss
  - Gender
  - Socioeconomic status

- **Variables that DO have predictive power:**
  - Cognitive skills
  - Age
  - Type of hearing loss (congenital vs acquired)



The difficulty of the lipreading task

Factors that increase difficulty include:

- Visibility of sounds
- Rapidity of speech
- Coarticulation and stress effects
- Visemes and homophenes
- Talker effects



Visibility of Sounds

- Most visible phonemes are made at the front of the mouth with maximum lip, jaw, and tongue movements
- *Ideal conditions 30% of speech is visible, with unusual conditions, 10-25% is visible*
- Some phonemes more visible than others
  - Consonants with lip closure
  - Consonants with tongue to front of mouth
- Vowels not considered visible (beak/book, bit/bought, boot/bet)
- Cannot see voiced vs voiceless

Rapidity of Speech

- Average 150 to 250 words per minute
- Speaking rate varies according to:
  - Emotion
  - Social etiquette
  - Fatigue
  - Articulation style

Coarticulation and Stress Effects

- Coarticulation
  - Same phoneme may look different
  - Influenced by vowel following consonant
- Same word looks different depending on speaker's stress
  - "What did ya do yesterday?" vs "What did YOU do yesterday?"

Visemes and Homophenes

- Visemes are groups of sounds that appear similar on the face
  - /b, m, p/
  - /k, g/
  - /n,d,t,s,z/
- Homophene: Any of a set of words that sound different, but look identical on a person's lips, so that they cannot be distinguished by lipreading.
  - (i.e. rise/rice, perch/merge, bind/mite, down/stout).

Talker effects

- The same sound can often look different when spoken by two or more different people.
- For example, the degree of mouth opening, or a person's accent can alter the look of sounds from person to person.

Factors Affecting Speechreading Performance

- How well a person can speech read is affected by at least four factors:
  - Speaker
  - Signal/Code
  - Environment
  - The Speech reader

- Figure 5.1

Speaker

- Should avoid shouting, mumbling, rapid speech, and turning away
- Easier to understand:
  - Clear, unexaggerated, slow speech (**clear speech)**
  - Familiar people
  - Native speakers
  - Facial expressions and hand gestures

Speaker's use of Clear Speech

- A way of speaking to enhance one's intelligibility; it entails speaking with a slowed rate and good but not exaggerated enunciation of words
  - "D'yeet yet?" for "Did you eat yet?"
  - "Go'n fishin' 'morrow af'ernoon" for "I'm going fishing tomorrow afternoon."

- Examples
  - An elementary school teacher
  - A newscaster
  - An airport employee who announces airline arrivals and departures
  - A public radio broadcaster
  - A politician
  - Queen Elizabeth

Signal/code

Structure of the message impacts speechreading. Easier to understand:

- Frequently used words
  - High frequency: words that occur frequently in everyday conversation (jersey)
  - Low frequency: words that occur infrequently in everyday conversation (cardigan)
- Words with few lexical neighbors
  - “bat” may be difficult to recognize, even though it begins with a highly visible mouth movement because many words look like it (e.g., bad, bet, mat, met, pat) and sound like it (e.g., cap, cat, scat, bad)
  - A word like “telephone” or “elephant” is easy because few look and sound like it
- Known linguistic context
  - Candace will buy the _____
  - Candace will set the _____ and chairs in the kitchen
  - Target word = table
- Known topic of message (e.g., “my summer vacation”)

Environmental Characteristics



Presence of background noise or reverberation impacts speechreading

Speechreading is affected by the talker's:

- Angle
  - 0 degrees azimuth=optimal
- Distance
  - 2-5 feet optimal
- Shadows on face cast by lighting

Speech reader

Speechreading is impacted by the speech reader's:

- Use of residual hearing
  - Even limited hearing ability can assist in speech recognition when combined with visual cues.
  - The importance of combining even some sound with visual cues shows why people with profound hearing loss are so dependent on hearing aids.
- Age and language skills
- Intelligence
- Degree and type of hearing loss
- Visual acuity
- Attention
- Attitude

Assessment of Speechreading

Standardized tests do exist

Table 5.3

Not used much at all

Why?

Problems with test validity

Problems with test difficulty

Problems with test, re-test reliability

Audiovisual Integration

- Speechreading is best used as a supplement to auditory information
- When both residual hearing and speechreading are available, the listener does much better on a communicative task (Fig 5-2)

Speechreading Training

Both Analytic and Synthetic

- Table 5.4
- **Analytic:** Before an entire word, sentence or phrase can be identified, it is necessary to perceive visually each if its basic parts
- **Synthetic:** Perception of the whole is paramount regardless of which parts are perceived visually

Speechreading Goal

- Underlying goal is to build the child's knowledge base concerning the speechreading process, and have the child develop appreciation of the benefits that speechreading provides.
- Stimuli should be client oriented
  - Real life situations
- Activities must be interesting and give the child opportunities to experience frequent success

Tips and tricks

- Watch the mouth
- Pay attention to facial expression
- Pay attention to gesture, posture, and movement
- Know the topic
- Attend to situational cues
- Keep informed
- Control anxiety and stress
- Use your hearing
- Try to ensure room is well-lit
- Minimize background noise
- Keep a positive attitude

Summary Remarks about Speechreading..

- Can speechreading skills be developed significantly through training?
- The jury is still out
- When improvements occur, they tend to be modest
- Sometimes we can’t objectively demonstrate improvement, but the client feels their skills are better
- Not much data available concerning children

Final Remarks

- Rarely is speechreading training provided without additional aural rehabilitation efforts.
- How patients combine the auditory and visual speech signals is a topic of great research.
- Findings will go a long way in the designing of more effective speechreading training protocols.



Table 5.5

TIPS FOR SPEECHREADING

Manual Communication

Types:

- Finger spelling
- ASL
- Manually Coded English
  - Signed English
  - Seeing Essential English (SEE II)
- Cued Speech

Manually Coded English

- Signed English

  Every word has a sign and then there are signs “markers” that reflect changes in verb form (past, present, future), number, possession, etc. In most cases then, you sign and word and add a marker at the end. Do “cook” and “cooks” and “cooked” and “cooking”.

- Seeing Essential English II (SEE II)

  Most current version of Signed English

American Sign Language (ASL)

- An independent language
- The only recognized language of the Deaf culture
- Unique vocabulary and syntax
- Aspects of prosody: facial expressions, head tilts, body movement, eye gazes
- Signs possess 4 identifying physical characteristics:
  - -hand configuration, movement, location, orientation

Cued Speech

Developed as a speechreading supplement to spoken English

- ✓ Provides visual cues for English phonemes not visible on the lips

Table 5.6

FORMS OF MANUAL AND SPOKEN COMMUNICATION